ISOMORPHIC INVARIANT ORDERS WITHOUT HERMITE EQUIVALENCE

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Abstract. Bhargava, Evertse, Győry, Remete and Swaminathan asked whether, in every degree $n \geq 4$, there are integral polynomials with isomorphic invariant orders that are not Hermite equivalent; they answered the question for $n = 4$. We answer it for every even n and for every n with $n-2$ divisible by 3, 5 or 7, with explicit pairs. In degree six one polynomial of the pair can be chosen monic.

Keywords: Hermite equivalence; invariant order; binary form.

MSC 2020: 11R04 (primary); 11C08, 11E16, 11R29.

1. Introduction

In 1857 Hermite introduced an equivalence relation on integral polynomials of a given degree n that is coarser than the classical $\mathrm{GL}_2(\mathbb{Z})$-equivalence, and proved that there are only finitely many classes of a given nonzero discriminant. Bhargava, Evertse, Győry, Remete and Swaminathan revived the notion and gave it an algebraic meaning [1]. To a polynomial

$$f(X) = f_0X^n + f_1X^{n-1} + \cdots + f_n \in \mathbb{Z}[X], \qquad f_0 \neq 0, \quad D(f) \neq 0,$$

one attaches its *invariant order* R_f and its *invariant ideal* I_f, a fractional ideal of R_f; both live in the étale algebra $K_f = \mathbb{Q}[X]/(f)$ and go back to Birch–Merriman, Nakagawa, Simon and Wood [5, 9, 10, 11]. For primitive f and g of nonzero discriminant, f and g are Hermite equivalent if and only if there is a ring isomorphism $R_f \cong R_g$ under which the ideal classes of I_f and I_g correspond [1, Corollary 3.12]. In degree three the condition on ideals is automatic, by the Levi–Delone–Faddeev correspondence, and for monic polynomials it is empty, since then $I_f = R_f$. This led the authors to the following question.[1]

Question (Bhargava–Evertse–Győry–Remete–Swaminathan)**.** Let $n \geq 4$. Do there exist (reducible or irreducible) polynomials $f, g \in \mathbb{Z}[X]$ of degree n and nonzero discriminant that have isomorphic invariant orders but are not Hermite equivalent, and if so, can such polynomials be exhibited?

In [1, Theorem 3.20] they exhibited two quartic polynomials with isomorphic invariant orders, one with principal and one with non-principal invariant ideal, which settles $n = 4$. For $n \geq 5$ the question has remained open. Our main result settles it in most degrees.

Theorem A. *Let $n \geq 5$ and suppose that n is even or that $n-2$ is divisible by 3, 5 or 7. Then there exist primitive polynomials $P, P' \in \mathbb{Z}[X]$ of degree n and the same nonzero discriminant whose invariant orders are isomorphic and which are not Hermite equivalent. For $n \geq 6$ there are infinitely many such pairs with pairwise distinct discriminants.*

Together with the quartic case, the question is answered for every $n \geq 4$ except the odd n for which $n-2$ is prime to 105; the first of these are $13, 15, 19, 21, 25$ (Figure 1). In the two smallest degrees the pairs are short enough to write down.

[1] Question 3.18 of [1]. We quote it in full.

Theorem B. *The sextic polynomials*

$$P(X) = X^6 - 18X^4 - 6X^3 + 79X^2 + 60X + 10,$$
$$P'(X) = 3X^6 + 8X^5 - 22X^4 - 82X^3 - 67X^2 - 20X - 2$$

have the same discriminant and isomorphic invariant orders, and they are not Hermite equivalent. The same holds for the quintic polynomials

$$Q(X) = 14X^5 - 19X^4 - 1676X^3 + 3714X^2 + 50843X - 152646,$$
$$Q'(X) = 42X^5 - 169X^4 - 2608X^3 + 12820X^2 + 975X - 45526.$$

The sextic P is monic. By [1, Corollary 3.16] two monic polynomials with isomorphic invariant orders are always Hermite equivalent, so Theorem B shows that this statement fails as soon as only one of the two polynomials is monic.

The idea behind both theorems is simple. Take two binary quadratic forms q_0 and q_1 of the same discriminant d, so that their invariant orders coincide, with I_{q_0} principal and I_{q_1} not. A single binary form H that is coprime to both q_0 and q_1 at every prime, in the sense that $\mathrm{Res}(q_0, H) = \mathrm{Res}(q_1, H) = \pm 1$, splits off a common factor: the invariant order of $q_i H$ is the product of the invariant orders of q_i and H, and the invariant ideal splits in the same way. The two products then have the same ring but invariant ideals that differ in exactly one factor, and a count of the principal factors shows that no isomorphism can identify them. The whole difficulty is to find such an H in the required degree. Section 5 shows that the obstruction lives in the Picard group of the quartic ring attached to $q_0 q_1$, and that one form of degree k produces forms in every degree divisible by k. Section 6 gives the four quadratic fields and the five forms that prove Theorem A.

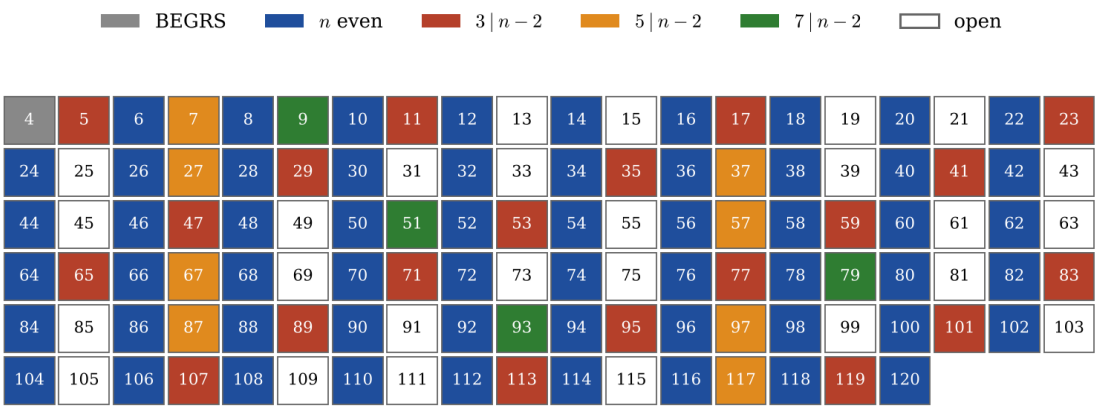


FIGURE 1. Degrees $4 \le n \le 120$ in which Question 3.18 of [1] is answered. Grey: the quartic example of [1]. Coloured: Theorem A, by the first divisibility condition it satisfies. White: degrees left open.

2. Invariant orders, invariant ideals and resultants

We identify a polynomial $f \in \mathbb{Z}[X]$ of degree n with the binary form $f(x, y) = y^n f(x/y)$ and use the same letter for both. Throughout, $f_0 = f(1, 0) \neq 0$ and $D(f) \neq 0$. Let α be the image of X in $K_f = \mathbb{Q}[X]/(f)$. The $\mathbb{Z}$-module $I_f(k) \subset K_f$, for $0 \le k \le n-1$, has basis

$$1,\ \alpha,\ \ldots,\ \alpha^k,\ \zeta_{k+1},\ \ldots,\ \zeta_{n-1}, \qquad \zeta_i = f_0\alpha^i + f_1\alpha^{i-1} + \cdots + f_{i-1}\alpha,$$

and $R_f = I_f(0)$, $I_f = I_f(1)$ [1, §3.1]. We use four facts from [1, Theorem 3.3], due to Birch–Merriman, Nakagawa, Simon and Wood:

(F1) R_f is an order in K_f and $D(R_f) = D(f)$.
(F2) $I_f(k)$ is a fractional ideal of R_f; it is invertible if and only if f is primitive, and then $I_f(k) = I_f^k$.
(F3) Let $S_f \subset \mathbb{P}^1_{\mathbb{Z}}$ be the closed subscheme cut out by f. Then R_f is the ring of global functions on S_f and $I_f(k)$ is the module of global sections of $\mathcal{O}(k)|_{S_f}$. Both are viewed inside K_f through the generic fibre $S_f \times \mathbb{Q} = \operatorname{Spec} K_f$, sections of $\mathcal{O}(k)$ being divided by y^k.
(F4) Primitive f, g of degree n and nonzero discriminant are Hermite equivalent if and only if there is a ring isomorphism $\varphi\colon R_f \to R_g$ with $\varphi(I_f)$ in the ideal class of I_g [1, Corollary 3.12].

Statement (F3) is Wood's geometric description [11, §2 and Appendix A].[2] The identification inside K_f uses that $f_0 \neq 0$, so the generic fibre of S_f lies in the chart $y \neq 0$, where it is $\operatorname{Spec} \mathbb{Q}[t]/(f(t,1))$; the section $x^j y^{k-j}$ goes to α^j.

For a binary quadratic form $q = ax^2 + bxy + cy^2$ of non-square discriminant $d = b^2 - 4ac$ with $a \neq 0$, let $\theta = (-b + \sqrt{d})/(2a)$. Then $K_q = \mathbb{Q}(\sqrt{d})$, $R_q = \mathbb{Z} + \mathbb{Z}a\theta = \mathbb{Z}[(d + \sqrt{d})/2]$ is the quadratic order $\mathcal{O}_d$ of discriminant d, and $I_q = \mathbb{Z} + \mathbb{Z}\theta$. In particular two quadratic forms of the same discriminant have literally the same invariant order. If q is primitive, then $aI_q = \mathbb{Z}a + \mathbb{Z}\frac{-b+\sqrt{d}}{2}$ is the invertible ideal of norm $|a|$ classically attached to q. If $a = 1$ then $I_q = R_q$ is principal.

For binary forms F, G of degrees $a, b \geq 1$ we write $\operatorname{Res}(F, G)$ for their resultant. We need the following standard properties [8, Chapter IV, §8 and Chapter IX, §3].

Lemma 2.1. *Let F, G, G_1, G_2, M be integral binary forms with $\deg F = a$, $\deg G = b$ and $\deg M = b - a \geq 0$.*

(1) $\operatorname{Res}(F, G) = \pm 1$ *if and only if F and G have no common zero in $\mathbb{P}^1(\overline{\mathbb{F}}_p)$ for any prime p.*
(2) $\operatorname{Res}(F, G_1G_2) = \operatorname{Res}(F, G_1)\operatorname{Res}(F, G_2)$ *and* $\operatorname{Res}(F, cG) = c^a \operatorname{Res}(F, G)$ *for $c \in \mathbb{Z}$.*
(3) $\operatorname{Res}(F, G + FM) = \operatorname{Res}(F, G)$.
(4) $D(FG) = D(F)D(G)\operatorname{Res}(F, G)^2$.

Proof. The resultant of binary forms is a universal integral polynomial in the coefficients, so it commutes with reduction modulo p, and over an algebraically closed field it vanishes exactly when the two forms have a common zero in $\mathbb{P}^1$. Since $\operatorname{Res}(F, G)$ is an integer, it equals ± 1 if and only if it is not divisible by any prime, which is (1). If $F = f_0 \prod_{i=1}^{a}(x - \alpha_i y)$ with $f_0 \neq 0$, then $\operatorname{Res}(F, G) = f_0^b \prod_i G(\alpha_i, 1)$ up to a sign depending only on a and b; this gives (2) and (3), because FM vanishes at every $(\alpha_i, 1)$. The general case of (2) and (3) follows by continuity in the coefficients, or by a change of variables in $\mathrm{GL}_2(\mathbb{Z})$ that makes $F(1, 0) \neq 0$. Part (4) is the usual product formula for discriminants. □

3. Splitting off a common factor

Lemma 3.1. *Let F and G be primitive binary forms of degrees $a, b \geq 2$ with nonzero leading coefficients $F(1, 0)$, $G(1, 0)$, nonzero discriminants, and $\operatorname{Res}(F, G) = \pm 1$. Then the isomorphism of $\mathbb{Q}$-algebras*

$$K_{FG} \longrightarrow K_F \times K_G, \qquad X \longmapsto (\alpha_F, \alpha_G),$$

[2] Wood works over an arbitrary base scheme and also treats forms of discriminant zero; we only need the case of integral forms with $f_0 D(f) \neq 0$, where S_f is finite and flat over $\operatorname{Spec} \mathbb{Z}$.

given by the Chinese remainder theorem maps R_{FG} onto $R_F \times R_G$ and I_{FG} onto $I_F \times I_G$.

Proof. On the affine chart $y \neq 0$ of $\mathbb{P}^1_{\mathbb{Z}}$ the schemes S_F, S_G and S_{FG} are cut out by $F(t,1)$, $G(t,1)$ and their product in $\mathbb{Z}[t]$. By Lemma 2.1(1) the scheme $\operatorname{Spec}\mathbb{Z}[t]/(F(t,1),G(t,1))$ has no points, so the two ideals are comaximal. For comaximal ideals the intersection equals the product, and the Chinese remainder theorem gives

$$\mathbb{Z}[t]/(F(t,1)G(t,1)) \cong \mathbb{Z}[t]/(F(t,1)) \times \mathbb{Z}[t]/(G(t,1)).$$

The same holds on the chart $x \neq 0$. Hence S_{FG} is the disjoint union of its closed subschemes S_F and S_G, and global sections of $\mathcal{O}(k)$ on S_{FG} are pairs of global sections on S_F and on S_G. On the generic fibre this decomposition is the map in the statement, since $\mathbb{Q}[t]/(F(t,1)G(t,1)) \cong \mathbb{Q}[t]/(F(t,1)) \times \mathbb{Q}[t]/(G(t,1))$ sends t to (α_F, α_G), and dividing by y^k commutes with it. Apply (F3) with $k = 0$ and $k = 1$; the latter is allowed because $a, b \geq 2$. □

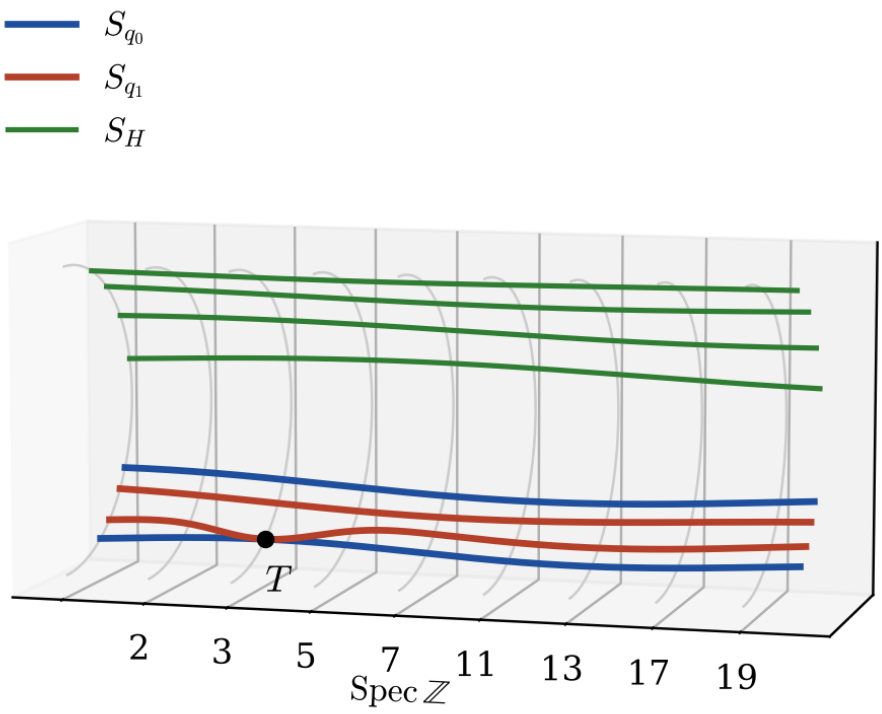


FIGURE 2. A schematic picture of $\mathbb{P}^1_{\mathbb{Z}} \to \operatorname{Spec}\mathbb{Z}$. The subschemes S_{q_0} and S_{q_1} of two quadratic forms of the same discriminant always meet, here over a single prime, in a finite scheme T. Theorem 4.1 needs a form H whose subscheme S_H avoids both of them in every fibre.

4. The twisting principle

Theorem 4.1. *Let q_0 and q_1 be primitive binary quadratic forms of the same non-square discriminant d with $q_0(1,0)q_1(1,0) \neq 0$, such that I_{q_0} is a principal ideal of $\mathcal{O}_d$ and I_{q_1} is not. Let H be a binary form of degree $k \geq 2$ with $H(1,0)D(H) \neq 0$ and*

$$\operatorname{Res}(q_0, H) = \pm 1, \qquad \operatorname{Res}(q_1, H) = \pm 1.$$

Then $P = q_0 H$ and $P' = q_1 H$ are primitive of degree $k+2$, $D(P) = D(P') = d\,D(H) \neq 0$, $R_P \cong R_{P'}$, and P and P' are not Hermite equivalent.

Proof. If a prime ℓ divided every coefficient of H, then ℓ^2 would divide $\operatorname{Res}(q_0, H)$ by Lemma 2.1(2). So H is primitive, and P, P' are primitive by Gauss's lemma. Their leading coefficients $q_i(1,0)H(1,0)$ are nonzero, and Lemma 2.1(4) gives $D(P) = D(P') = d\,D(H)$, which is nonzero.

By Lemma 3.1, applied to (q_0, H) and to (q_1, H),

$$R_P \cong \mathcal{O}_d \times R_H \cong R_{P'}, \qquad I_P \leftrightarrow I_{q_0} \times I_H, \qquad I_{P'} \leftrightarrow I_{q_1} \times I_H.$$

Suppose that P and P' were Hermite equivalent. By (F4) there is a ring isomorphism $\varphi\colon R_P \to R_{P'}$ and $\kappa \in K_{P'}^{\times}$ with $\varphi(I_P) = \kappa I_{P'}$.

The order R_H lies in the reduced algebra K_H, so it has finitely many idempotents, and it is the product of the rings eR_H over its primitive idempotents e; each eR_H has no idempotents other than 0 and e. The primitive idempotents of $R_P = \mathcal{O}_d \times R_H$ are $(1,0)$ and the $(0,e)$, because $\mathcal{O}_d$ is a domain. A ring isomorphism sends primitive idempotents bijectively to primitive idempotents, and for each primitive idempotent ε of R_P it restricts to an isomorphism $\varepsilon R_P \to \varphi(\varepsilon)R_{P'}$ carrying εI_P onto $\varphi(\varepsilon)\kappa I_{P'}$, a principal multiple of $\varphi(\varepsilon)I_{P'}$.

For a ring A with an ideal J write $N(A, J)$ for the number of primitive idempotents ε of A such that εA is isomorphic to $\mathcal{O}_d$ and εJ is a principal ideal of εA. The previous paragraph shows $N(R_P, I_P) = N(R_{P'}, I_{P'})$. On the other hand the component $(1,0)$ contributes 1 to $N(R_P, I_P)$, because I_{q_0} is principal, and 0 to $N(R_{P'}, I_{P'})$, because I_{q_1} is not; whether an ideal of $\mathcal{O}_d$ is principal does not depend on the choice of isomorphism, since $\mathcal{O}_d$ has only the identity and conjugation as automorphisms. The components $(0,e)$ contribute the same amount to both counts, because they carry the same pair (eR_H, eI_H). Hence $N(R_P, I_P) = N(R_{P'}, I_{P'}) + 1$, a contradiction. □

Remark 4.2. The proof uses only that I_{q_0} and I_{q_1} lie in different orbits of $\operatorname{Pic}(\mathcal{O}_d)$ under conjugation. The principal case is the most convenient, because then q_0 can be taken monic.

5. Finding the common factor

Fix q_0, q_1 as in Theorem 4.1 and call a binary form H *admissible* if $\operatorname{Res}(q_0, H) = \pm 1$ and $\operatorname{Res}(q_1, H) = \pm 1$. Admissibility is stable under products and under adding multiples of $q_0 q_1$.

Proposition 5.1. *Let $H_1, \dots, H_s$ be admissible forms of degrees $k_1, \dots, k_s$, let $m_1, \dots, m_s \geq 0$, and suppose $k = \sum m_j k_j \geq 4$. Put $M = \prod_{i=1}^{k-4}(x - iy)$ and*

$$H_t = \prod_{j=1}^{s} H_j^{m_j} + t\, q_0 q_1 M \qquad (t \in \mathbb{Z}).$$

Then every H_t is admissible of degree k, and $H_t(1,0)D(H_t) \neq 0$ for all but finitely many t. The values $D(H_t)$ are not all equal.

Proof. Admissibility of H_t follows from Lemma 2.1(2) and (3), since q_0q_1M is divisible by q_0 and by q_1. The leading coefficient $H_t(1,0)$ is a polynomial of degree one in t with slope $q_0(1,0)q_1(1,0) \neq 0$. The discriminant is homogeneous of degree $2k-2$ in the coefficients, so $D(H_t)$ is a polynomial in t of degree at most $2k-2$ whose coefficient of t^{2k-2} is

$$D(q_0q_1M) = D(q_0)\,D(q_1)\,D(M)\,\operatorname{Res}(q_0,q_1)^2\,\operatorname{Res}(q_0,M)^2\,\operatorname{Res}(q_1,M)^2.$$

This is nonzero: q_0 and q_1 are irreducible over $\mathbb{Q}$ and not proportional, because their invariant ideals are not in the same class; M splits into distinct rational linear factors. Hence $D(H_t)$ is a nonconstant polynomial in t, which vanishes for finitely many t only and takes infinitely many values. □

The next statement explains when admissible forms exist. It is not needed for the proof of Theorem A, but it is how the forms in Section 6 were found, and it shows what goes wrong in the remaining degrees.

Proposition 5.2. *Let $F = q_0q_1$ and $k \geq 3$. An admissible form of degree k exists if and only if I_F^k is a principal ideal of the quartic order R_F.*

Proof. By Lemma 2.1(1),(2) a form H of degree k is admissible if and only if H has no zero on S_F, that is, if and only if the section $H|_{S_F}$ of the invertible sheaf $\mathcal{O}(k)|_{S_F}$ vanishes nowhere. Since S_F is affine, this holds if and only if $H|_{S_F}$ generates the module of global sections, which is I_F^k by (F2) and (F3); for $k \geq 4$ one uses that $\mathcal{O}(k)|_{S_F}$ is the k-th tensor power of the invertible sheaf $\mathcal{O}(1)|_{S_F}$, whose module of sections I_F is invertible, so that the sections form $I_F^{\otimes k} = I_F^k$. So an admissible H makes I_F^k principal. Conversely, the sequence $0 \to \mathcal{O}(k-4) \xrightarrow{F} \mathcal{O}(k) \to \mathcal{O}(k)|_{S_F} \to 0$ on $\mathbb{P}^1_{\mathbb{Z}}$ and the vanishing of $H^1(\mathbb{P}^1_{\mathbb{Z}}, \mathcal{O}(k-4))$ for $k \geq 3$ [7, Theorem III.5.1] show that every global section of $\mathcal{O}(k)|_{S_F}$ is the restriction of a form of degree k. A generator of I_F^k is therefore the restriction of an admissible form. □

The class of I_F maps to $(1, [I_{q_1}])$ under $\operatorname{Pic}(R_F) \to \operatorname{Pic}(\mathcal{O}_d)^2$, so I_F^k can be principal only if $[I_{q_1}]^k = 1$. The kernel of that map is $\mathcal{O}_T^\times$ modulo the images of the two unit groups, where $\mathcal{O}_T$ denotes the ring of functions on the finite scheme $T = S_{q_0} \cap S_{q_1}$ (Figure 2). The order of I_F can therefore be larger than that of $[I_{q_1}]$, and it depends on the chosen forms within their classes; a large unit group helps, which is why all our examples come from real quadratic fields. Since an admissible form of degree $n-2$ forces $[I_{q_1}]^{n-2} = 1$ while $[I_{q_1}] \neq 1$, an odd degree n always needs a quadratic order whose class group has an element of odd order dividing $n-2$.

6. Four quadratic fields

Table 1 lists four pairs (q_0, q_1). In each case q_0 is monic, so $I_{q_0} = \mathcal{O}_d$ is principal. The discriminant d is fundamental, $\mathcal{O}_d$ is the full ring of integers of $\mathbb{Q}(\sqrt{d})$, and the class number h is given for orientation only.

d	h	q_0	q_1	admissible base forms
40	2	$x^2 - 10y^2$	$2x^2 + 4xy - 3y^2$	H_4, H_6
229	3	$x^2 + xy - 57y^2$	$3x^2 - 5xy - 17y^2$	H_3
401	5	$x^2 + xy - 100y^2$	$2x^2 + 23xy + 16y^2$	H_5
577	7	$x^2 + xy - 144y^2$	$2x^2 - 11xy - 57y^2$	H_7

Table 1. The quadratic data. The base forms are listed in the appendix.

Lemma 6.1. *In each row of Table 1 the ideal I_{q_1} is not principal.*

Proof. Write $q_1 = ax^2 + bxy + cy^2$. By Section 2, $aI_{q_1} = \mathbb{Z}a + \mathbb{Z}\frac{-b+\sqrt{d}}{2}$ is an ideal $\mathfrak{p}$ of norm a, which is the prime 2 or 3. If $\mathfrak{p}$ were principal, $\pm a$ would be the norm of an element of $\mathcal{O}_d$ not divisible by any rational integer, so $\pm a$ would be primitively represented by the principal form q_0.

For $d = 40$ this is impossible modulo 5: $x^2 - 10y^2 \equiv x^2 \pmod 5$, and ± 2 are not squares modulo 5.

For the other three rows we use reduction theory of indefinite forms [6, Chapter 3]. A form (A, B, C) of discriminant $d > 0$ is reduced if $0 < B < \sqrt{d}$ and $\sqrt{d} - B < 2|A| < \sqrt{d} + B$; two reduced forms are equivalent if and only if they lie on the same cycle. If m is primitively represented by q_0 and $|m| < \sqrt{d}/2$, then q_0 is equivalent to a form (m, B, C), and replacing B by its representative modulo $2|m|$ in the interval $(\sqrt{d} - 2|m|, \sqrt{d})$ makes it reduced. So $\pm a$ would appear as a first coefficient on the cycle of q_0. Here $a < \sqrt{d}/2$ in all three cases ($\sqrt{d}/2 > 7.5$), and the cycles of the principal forms are

$$d = 229: \quad (1, 15, -1),\ (-1, 15, 1);$$

$$d = 401: \quad (1, 19, -10),\ (-10, 1, 10),\ (10, 19, -1),\ (-1, 19, 10),\ (10, 1, -10),\ (-10, 19, 1);$$

$$d = 577: \quad (1, 23, -12),\ (-12, 1, 12),\ (12, 23, -1),\ (-1, 23, 12),\ (12, 1, -12),\ (-12, 23, 1).$$

Each step $(A, B, C) \mapsto (C, B', C')$ chooses $B' \equiv -B \pmod{2|C|}$ in $(\sqrt{d} - 2|C|, \sqrt{d})$, and one checks $B'^2 - 4CC' = d$ directly. The first coefficients are ± 1 for $d = 229$ and $\pm 1, \pm 10$, resp. $\pm 1, \pm 12$, for $d = 401$, 577. Neither ± 3 nor ± 2 occurs. □

Lemma 6.2. *The forms H_4, H_6 (for $d = 40$), H_3 (for $d = 229$), H_5 (for $d = 401$) and H_7 (for $d = 577$) listed in the appendix are admissible for the corresponding pair (q_0, q_1), have nonzero leading coefficient and nonzero discriminant, and are irreducible over $\mathbb{Q}$.*

Proof. For each base form H and each i the number $\mathrm{Res}(q_i, H)$ is the determinant of an integer Sylvester matrix of size $\deg H + 2$; equivalently, with $q_i = a_i x^2 + b_i xy + c_i y^2$ and $\theta_i = (-b_i + \sqrt{d})/(2a_i)$, it equals $a_i^{\deg H} N_{\mathbb{Q}(\sqrt{d})/\mathbb{Q}}(H(\theta_i, 1))$. These finite computations give the values in the appendix, all equal to ± 1. Irreducibility and $D(H) \neq 0$ are again finite checks; irreducibility is not used below. □

Proof of Theorem A. Let $n \geq 5$ and $k = n - 2 \geq 3$. If n is even, write $k = 4a + 6b$ with $a, b \geq 0$ and use the row $d = 40$ with H_4, H_6. If $3 \mid k$, $5 \mid k$ or $7 \mid k$, use the row $d = 229$, 401 or 577 with the single base form H_3, H_5 or H_7 and exponent $k/3$, $k/5$ or $k/7$. If k equals the degree of the base form, take H to be that form; it is admissible with $H(1, 0)D(H) \neq 0$ by Lemma 6.2. Otherwise $k \geq 4$, and Proposition 5.1 gives admissible H_t of degree k with $H_t(1, 0)D(H_t) \neq 0$ for all but finitely many t, and with infinitely many distinct discriminants. By Lemma 6.1, Theorem 4.1 applies to $P = q_0 H$ and $P' = q_1 H$. Their common discriminant $d\, D(H_t)$ takes infinitely many values as t varies, which proves the last assertion; $k \geq 4$ holds for every $n \geq 6$. □

Proof of Theorem B. For the sextic pair take $d = 40$, $q_0 = x^2 - 10y^2$, the form $q_1^* = 3x^2 + 8xy + 2y^2$ of discriminant 40, and $H = x^4 - 8x^2y^2 - 6xy^3 - y^4$. Then $P = q_0 H$ and $P' = q_1^* H$, and one computes $\mathrm{Res}(q_0, H) = \mathrm{Res}(q_1^*, H) = 1$ and $D(H) \neq 0$. The ideal $3I_{q_1^*} = \mathbb{Z}3 + \mathbb{Z}(-4 + \sqrt{10})$ has norm 3, and $x^2 - 10y^2 = \pm 3$ has no solution modulo 5, so $I_{q_1^*}$ is not principal. Theorem 4.1 applies. For the quintic pair take the row $d = 229$ of Table 1 and $H = H_3$; then $Q = q_0 H_3$ and $Q' = q_1 H_3$. □

7. Concluding remarks

Remark 7.1. The forms found here come with a great deal of freedom. Proposition 5.1 produces, in each admissible degree $k \geq 4$, an arithmetic progression of pairs, and different choices of q_1 in its $\mathrm{SL}_2(\mathbb{Z})$-orbit, or of the auxiliary form M, give further families. By Proposition 5.2, the existence of an admissible form of degree k depends only on the class of $I_{q_0 q_1}^{k}$, so the search for base forms reduces to a computation in $\mathcal{O}_T^{\times}$ modulo units.

Remark 7.2. The remaining odd degrees, those with $\gcd(n-2, 105) = 1$, need for some prime $\ell \mid n-2$ a quadratic order with an ideal class of order ℓ together with a pair (q_0, q_1) for which $I_{q_0 q_1}^{\ell}$ is principal. Real quadratic fields with class number divisible by ℓ exist for every ℓ, and for each fixed ℓ the condition is a finite computation, so every further prime ℓ adds the residue class $n \equiv 2 \pmod{\ell}$. A proof covering all odd n at once would need control over the unit group of $\mathcal{O}_d$ modulo the intersection scheme T in a family of fields, which we do not have.

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Declarations

Data availability. All data needed to verify the results are contained in the paper and its appendix. The LaTeX source, the figure script and a script that recomputes the resultants of the appendix are archived at doi:10.5281/zenodo.XXXXXXXX.

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Appendix A. Base forms

A form $h_0 x^k + h_1 x^{k-1} y + \cdots + h_k y^k$ is written as its coefficient vector $[h_0, \ldots, h_k]$. The last two columns give the resultants with q_0 and q_1 of Table 1.

d	form	coefficients	$\mathrm{Res}(q_0, \cdot)$	$\mathrm{Res}(q_1, \cdot)$
40	H_4	$[-213, 254, 1679, -1135, 67]$	-1	1
40	H_6	$[-1, -2, 11, 19, -9, 9, -7]$	-1	1
229	H_3	$[14, -33, -845, 2678]$	1	-1
401	H_5	$[526505, 10598024, 47546124, -52212324,$ $69923173, -93789359]$	-1	1

For $d = 577$ the form $H_7 = [h_0, \ldots, h_7]$ has $\mathrm{Res}(q_0, H_7) = -1$ and $\mathrm{Res}(q_1, H_7) = 1$, with

$$\begin{aligned}
h_0 &= 923365420800961134849136086966499 9,\\
h_1 &= -158785974126310386817980985635657596,\\
h_2 &= 351336977243533453505499571533318677,\\
h_3 &= 295684110654525425618811641878008 2341,\\
h_4 &= -535022898010655232998946727047265289,\\
h_5 &= 228203582350855930536981792812157305,\\
h_6 &= -625031829845431344394388625520027 51,\\
h_7 &= 202185608478454766021956515566341 69.
\end{aligned}$$

The polynomials of Theorem B are $P = (x^2 - 10y^2)(x^4 - 8x^2y^2 - 6xy^3 - y^4)$, $P' = (3x^2 + 8xy + 2y^2)(x^4 - 8x^2y^2 - 6xy^3 - y^4)$, $Q = (x^2 + xy - 57y^2)H_3$ and $Q' = (3x^2 - 5xy - 17y^2)H_3$, written as polynomials in $X = x/y$.

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